

Artistic Statement Update

Conceptual Overview

The application is a web-based sentiment analysis tool designed to provide real-time emotional support. Combining facial expression detection and sentiment analysis of speech, it creates a supportive digital companion experience. By bridging AI capabilities with empathetic response generation, the app enhances emotional resilience and user engagement.

Future Integration

- **Enhanced Models:** Incorporate multimodal transformers for deeper emotional understanding.
- **Voice-Based Responses:** Add audio output to improve user interaction.
- **Gaming Integration:** Expand functionality to integrate directly with gaming platforms for enhanced immersion.

Use of AI Tools

- **Hugging Face Models:**
 - dima806/facial_emotions_image_detection: Facial sentiment analysis.
 - openai/whisper-small: Speech-to-text transcription.
 - cardiffnlp/twitter-roberta-base-sentiment-latest: Text sentiment analysis.
- **OpenAI GPT-4:** Empathetic text generation.
- **Frontend Frameworks:**
 - p5.js and HTML for video/audio inputs.
- **Backend Framework:**
 - Flask for API management and model integration.